

PURBANCHAL UNIVERSITY
FACULTY OF ENGINEERING
Course Structure of Bachelor in Bio-Medical Engineering
Effective from 2021 Batch

First Semester										
S.N.	Subjects	C.H.	L	T	P	Internal		Final		Total
						Th.	P.	Th.	P.	
1	Mathematics I	3	3	3	-	40	-	60	-	100
2	Physics	3	4	2	2	40	10	60	15	125
3	Computer Programming	3	3	1	3	40	30	60	20	150
4	Basic Electrical Engineering	2	2	1	1	20	25	30	-	75
5	Basic Electronics Engineering	3	3	1	3/2	40	25	60	-	125
6	Engineering Drawing	3	1	-	3	-	60	-	40	100
	Total	17	16	8	10.5					675

Second Semester										
S.N.	Subjects	C.H.	L	T	P	Internal		Final		Total
						Th.	P.	Th.	P.	
1	Mathematics II	3	3	3	-	40		60	-	100
2	Chemistry	3	3	2	2	40	10	60	15	125
3	Basic Mechanical Engineering	3	3	1	3	40	25	60	-	125
4	Electro Engineering Materials	3	3	1	-	40	25	60	-	125
5	Introduction to Computational System	3	3	1	3	40	25	60	-	125
6	Digital Electronics	3	3	1	3	40	25	60	-	125
	Total	18	18	9	11					725

Note: C.H.: Credit Hour L: Lecture T: Tutorial P: Practical Th: Theory
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Third Semester										
S.N.	Subjects	C.H.	L	T	P	Internal		Final		Total
						Th.	P.	Th.	P	
1	Mathematics III	3	3	3	-	40		60	-	100
2	Bio-Engineering Materials and Components	4	4	1	-	40	-	60	-	100
3	Human Anatomy and Physiology I	4	4	1	3	40	25	60	-	125
4	Fluid Mechanics & Thermodynamics	3	3	1	-	40	-	60	-	100
5	Cell Biology and Immunology	3	3	1	3	40	25	60	-	125
6	Microprocessor	3	3	1	3	40	50	60	-	150
	Total	20	20	8	9					700

Fourth Semester										
S.N.	Subjects	C.H.	L	T	P	Internal		Final		Total
						Th.	P.	Th.	P	
1	Applied Mathematics	3	3	3	-	40		60	-	100
2	Applied Sociology	3	3	1	-	40		60	-	100
3	Electronic Devices and Circuits	3	3	1	3/2	40	25	60	-	125
4	Biomechanics	4	4	1	-	40	-	60	-	100
5	Biomedical Embedded System Design	3	3	1	3	40	25	60	-	125
6	Human Anatomy and Physiology II	4	4	1	3	40	25	60	-	125
	Total	20	20	8	7.5					675

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Fifth Semester										
S.N.	Subjects	C.H.	L	T	P	Internal		Final		Total
						Th.	P.	Th.	P	
1	Probability and Statistics	3	3	1	-	20	20	60	-	100
2	Numerical Methods	3	3	1	2	40	30	60	20	150
3	Implantable Devices	3	3	1	2	40	-	60	-	100
4	Control Systems	3	3	1	3	40	10	60	15	125
5	Tissue Device Interactions	3	3	1	-	40	-	60	-	100
6	Communication Systems	3	3	1	2	40	25	60	-	125
	Total	18	18	6	9					700

Sixth Semester										
S.N.	Subjects	C.H	L	T	P	Internal		Final		Total
						Th.	P.	Th.	P	
1	Engineering Economics	3	3	1	-	40	-	60	-	100
2	Medical Industry Management	3	3	1	-	40	-	60	-	100
3	Medical Imaging I	3	3	1	2	40	25	60	-	125
4	Biomedical Instrumentation I	4	4	1	2	40	25	60	-	125
5	Medical Electronics	3	3	1	2	40	25	60	-	125
6	Biomedical Digital Signal Processing	3	3	1	2	40	25	60	-	125
	Total	19	19	6	8					700

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Seventh Semester										
S.N.	Subjects	C.H.	L	T	P	Internal		Final		Total
						Th.	P.	Th.	P	
1	Organization & Project Management	3	3	1	-	40		60	-	100
2	Biomedical Instrumentation II	4	4	1	2	40	25	60	-	125
3	Medical Imaging II	3	3	1	2	40	25	60	-	125
4	Elective I	3	3	1	2	40	25	60	-	125
5	Elective II	3	3	1	2	40	25	60	-	125
6	Project	6	-	3	9	-	120	-	80	200
	Total	22	22	5	14					800

Eighth Semester										
S.N.	Subjects	C.H.	L	T	P	Internal		Final		Total
						Th.	P.	Th.	P	
1	Engineering Professional Practice	2	2	1	-	20	-	30	-	50
2	Elective III	3	3	1	2	40	25	60	-	125
3	Internship	3	3	1	-	-	60	-	40	100
4	Project	6	-	3	9	-	120	-	80	200
	Total	14	14	3	8					475

Total

150

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Elective I

1	Structural & Functional Biomaterials	3	3	1	-	4
2	Biomedical Equipment Maintenance I	3	3	1	2	6
3	Physiological Modeling	3	3	1	2	6

Elective II

1	Minimally Invasive Medical Technology	3	3	1	-	4
2	Medical Image Processing	3	3	1	2	6
3	Principles of Tissue Engineering	3	3	1	-	4
4	Theory of Medical Robotics	3	3	1	2	6

Elective III

1	Neural Network	3	3	1	2	6
2	Medical Informatics	3	3	1	2	6
3	Telemedicine and Telehealth	3	3	1	-	4
4	Biomedical Applications of Nanotechnology	3	3	1	-	4
5	Biomedical Equipment Maintenance II	3	3	1	2	6
6	Bio-Electromagnetism	3	3	1	2	6